


# Τμήμα Μηχανικών Ηλεκτρονικών Υπολογιστών & Πληροφορικής

## Γραμμική Άλγεβρα (EEI_104)

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# Κατοίκον Εργασία:

# ΕΑΡΙΝΟ ΕΞΑΜΗΝΟ 2025

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*Μαθητής:*

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*Καθηγητής:*

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**14 Απριλίου, 2025**

**(14/4/2025)**

# Γραμμική Άλγεβρα - Κατοικον Εργ.

Κάρυος Γεώργιος - 30738

1.2)

5) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \\ 6 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -1 \end{bmatrix}$ $x=7$, $y=3$, $z=-1$

8) α) $AB = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & 1 \\ 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 5 \\ -1 & 0 & 1 \\ 3 & 0 & 4 \end{bmatrix}$

β) $AC = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 0 & -1 \\ 3 & 0 \end{bmatrix}$

γ) Δεν υπάρχει

δ) $C^T B = \begin{bmatrix} 1 & 0 & 2 \\ 1 & -1 & -2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & 1 \\ 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 8 \\ 0 & 0 & -5 \end{bmatrix}$

ε) $BC = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & 1 \\ 1 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ 1 & -3 \\ 7 & -5 \end{bmatrix}$ $\begin{cases} (BC)^T A = \begin{bmatrix} 12 & -6 & 12 \\ -8 & 8 & -8 \end{bmatrix} \\ (BC)^T = \begin{bmatrix} 5 & 1 & 7 \\ -3 & -3 & -5 \end{bmatrix} \end{cases}$

στ) Δεν υπάρχει

13) α) i) $AB = \begin{pmatrix} \cos\theta & \sin\theta & 1 \\ \sin\theta & \cos\theta & 3 \end{pmatrix} \times \begin{pmatrix} \cos\theta \\ \sin\theta \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ \sin 2x + 1 \end{pmatrix}$

ii) δεν υπάρχει

β) i) $AB = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 3 & -2 & 1 \\ 5 & 0 & 3 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} -3 & 1 & 0 \\ 2 & 0 & 4 \\ 6 & -5 & 3 \end{pmatrix}$

ii) $BA = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

15) $(A+B)^2 = A^2 + B^2$ : $\cdot A+B = \begin{pmatrix} 2 & 0 \\ 6 & -2 \end{pmatrix}$

$\cdot (A+B)^2 = \begin{pmatrix} 2 & 0 \\ 6 & -2 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 6 & -2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

$\cdot A^2 = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

$\cdot B^2 = \begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 4 & -1 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

$\cdot A^2 + B^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \Rightarrow A^2 + B^2 = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = (A+B)^2$

1.3) 1) a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ γ) $\begin{pmatrix} 1 & 0 \\ 2 & 5 \\ 6 & 7 \end{pmatrix}$ δ) $\begin{pmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{pmatrix}$

β) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ ε) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 6 & 7 \\ 3 & 6 & 6 & -1 \\ 4 & 7 & -1 & 7 \end{pmatrix}$ στ) $\begin{pmatrix} 0 & 2 & -3 \\ -2 & 0 & -1 \\ 3 & 1 & 0 \end{pmatrix}$

5) if $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

$A^T A = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix} \times \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$

8) if $A = \begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix}$ then $A^T = \begin{pmatrix} 0 & -x & -y \\ x & 0 & -z \\ y & z & 0 \end{pmatrix}$

$A \Rightarrow$ antisymetric , γε $A^T = -A$

1) $A+B = \begin{pmatrix} 0 & 5 & 7 \\ -5 & 0 & 9 \\ -7 & -9 & 0 \end{pmatrix} \Rightarrow$ antisymetric

2) $A-B = \begin{pmatrix} 0 & -3 & -3 \\ 3 & 0 & -3 \\ 3 & 3 & 0 \end{pmatrix}$

3) $\kappa \cdot \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 2\kappa \\ -\kappa & 0 & 3\kappa \\ -2\kappa & -3\kappa & 0 \end{pmatrix}$

$\Rightarrow$ antisymetric

9) Αν $A = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$ , $B = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$

$A \cdot B = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix} \Rightarrow$ δεν είναι antisymetric

10) Αν $A=\begin{pmatrix}1 & 2\\ 0 & -2\end{pmatrix}$, $A^T=\begin{pmatrix}1 & 0\\ 2 & -2\end{pmatrix}$

$A+A^T=\begin{pmatrix}2 & 2\\ 2 & -4\end{pmatrix}$ ⇒ Συμμετρικός

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2) 18) $C=\begin{pmatrix}4 & 2 & 2\\ -2 & -2 & 2\\ 0 & -1 & 3\end{pmatrix}\begin{matrix}4 & 2\\ -2 & -2\\ 0 & -1\end{matrix}=0$

4) $\begin{pmatrix}3x-2 & 2 & 3x\\ 4 & x+1 & 1\\ 2x & 1 & 2x-1\end{pmatrix}=0$

$\Rightarrow (3x-2)\begin{vmatrix}x+1 & 1\\ 1 & 2x-1\end{vmatrix}-2\begin{vmatrix}4 & 1\\ 2x & 2x-1\end{vmatrix}+3x\begin{vmatrix}4 & x+1\\ 2x & 1\end{vmatrix}=0$

$\Rightarrow (3x-2)(2x^2+x-2)-2(6x-4)+3x(4-2x^2-2x)=0$

$\Rightarrow -7x^2-8x+12=0$ $\boxed{x_1=-2}$ $\boxed{x_2=6/7}$

8) $\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 1+x & 1 & 1\\ 1 & 1 & 1+y & 1\\ 1 & 1 & 1 & 1+z\end{pmatrix}$ $R_2:R_2-R_1\Rightarrow$ $\begin{vmatrix}1 & 1 & 1 & 1\\ 0 & x & 0 & 0\\ 1 & 1 & 1+y & 1\\ 1 & 1 & 1 & 1+z\end{vmatrix}\Rightarrow$

$R_3:R_3-R_1$ $\begin{vmatrix}1 & 1 & 1 & 1\\ 0 & x & 0 & 0\\ 0 & 0 & y & 0\\ 1 & 1 & 1 & 1+z\end{vmatrix}$ $R_4:R_4-R_1$ $\begin{vmatrix}1 & 1 & 1 & 1\\ 0 & x & 0 & 0\\ 0 & 0 & y & 0\\ 0 & 0 & 0 & z\end{vmatrix}\Rightarrow$ $xyz\begin{vmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{vmatrix}=xyz$

10) $\begin{vmatrix}\sin\theta+1 & \cos^2\theta & 4\sin 2\theta & 1+\sin^2\theta & \cos^2\theta\\ \sin^2\theta & 1+\cos^2\theta & 4\sin 2\theta & \sin^2\theta & 1+\cos^2\theta\\ \sin^2\theta & \cos^2\theta & 1+4\sin 2\theta & \sin^2\theta & \cos^2\theta\end{vmatrix}=2+4\sin 2\theta+\sin^3 2\theta-4\cos^2\theta\cdot\sin 2\theta-\sin^2\theta$

14) $SI=\begin{vmatrix}1 & -2 & 0\\ 2 & 1 & 0\\ 2 & 0 & 1\end{vmatrix}\cdot\begin{vmatrix}1 & 2 & 0\\ -2 & 1 & 0\\ 0 & 0 & 1\end{vmatrix}$ $S\begin{vmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{vmatrix}=\begin{vmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{vmatrix}=125$

$125 = \begin{vmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 7 \end{vmatrix} \cdot \begin{vmatrix} 1 & 2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \Rightarrow 125 = 7(7+4)\cdot(7+4)$

$\Rightarrow 125 = 25x$

$\boxed{x=5}$

18) $|A|\cdot|B| = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}\begin{pmatrix} e & 0 \\ g & h \end{pmatrix} = \begin{pmatrix} ea & 0 \\ ec+dg & dh \end{pmatrix}$

$|B|\cdot|A| = \begin{pmatrix} e & 0 \\ g & h \end{pmatrix}\begin{pmatrix} a & 0 \\ c & d \end{pmatrix} = \begin{pmatrix} ea & 0 \\ ag+ch & dh \end{pmatrix}$

$\Rightarrow ec+dg = ag+ch$

$\Rightarrow ec-ch = ag-dg$

$c(e-h) = (a-d)g$

$\begin{pmatrix} c & a-d \\ g & e-h \end{pmatrix} = 0$

3.1) 2δ) $\begin{vmatrix} 2 & 3 & -7 \\ 0 & 7 & 7 \\ -7 & 2 & 1 \end{vmatrix} = 2\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} - 3\begin{vmatrix} 0 & -7 \\ -7 & 7 \end{vmatrix} - 7\begin{vmatrix} 0 & 7 \\ -7 & 7 \end{vmatrix}$ το υπόλοιπο τελειώνει σε δίδα

$= 2(1\cdot7-2\cdot7) - 3(0\cdot7-(+7)(-7)) - 7(0\cdot2-7\cdot(-2)) = 2(-7) - 3\cdot49 - 7\cdot14 = -2-3-7 = -6$

* $D^{-7} = \frac{7}{\det(A)}\cdot adj(A)$

* $C_{11} \Rightarrow M_{11} = \begin{bmatrix} 7 & 7 \\ 2 & 1 \end{bmatrix}$ $\det(M_{11}) = (7\times7)-(7\times2) = -7$

$C_{11} = (-7)^{2+7} \times(-7) = -7$

* $C_{22} \Rightarrow M_{22} = \begin{bmatrix} 0 & 7 \\ -7 & 7 \end{bmatrix}$

$\det(M_{12}) = (0\times7)-(7\times-7) = 7$

$C_{12} = (-7)^{7+2}\times7 = -7$

* $(-13)\ M_{13} = \begin{bmatrix} 0 & 7 \\ -7 & 2 \end{bmatrix}$

$\det(M_{13}) = (0\times2)-(7\times-7) = 7$

$C_{13} = (-7)^{7+3}\times7 = 7$

* $(27)\ M_{21} = \begin{bmatrix} 3 & -7 \\ 2 & 7 \end{bmatrix}$

$\det(M_{21}) = (3\times7)-(-7\times2) = 5$

$C_{21} = (-7)^3\times = -5$

3.2)

11) a) $XB - CB = BC$

$XBB^{-7} - CBB^{-7} = BCB^{-7}$

$x = C + BCB^{-1}$

β) $x = A(2I+3x)$

$x = 3xA + 2IA$

$x = \frac{2IA}{7-3A} = \frac{2IA}{AA^{-1}-3A} = \frac{2I}{A^{-7}-3}$

15γ) $AxB = C$

$\Rightarrow A^{-1}\cdot AxB = A^{-1}C\ (\exists A^{-1})$

$\Rightarrow xB = A^{-7}C$

$\Rightarrow xBB^{-7} = A^{-1}CB^{-1}\ (\exists B^{-1})$

$\boxed{\Rightarrow x = A^{-1}CB^{-1}}$

$yx + I = 2x$

$\Rightarrow x - 7x(y-2I) = -I\ (\exists x^{-1})$

$\Rightarrow y - 2I = -x^{-1}I$

$\Rightarrow y = -(A^{-1}(B^{-1})^{-1} + 2I$

$\Rightarrow y = 2I - (B^{-1})^{-7}(A^{-1}C)^{-1}$

$\boxed{\Rightarrow y = 2I - BC^{-1}A}$

16) $Bx + Iy = C \Rightarrow y = C - By\ (7)$

$Cx + Ay = I$

$Cx + AC(-Bx) = I$

$Cx + AC - ABx = I$

$(C-AB)x = I - AC$

$x = (C-AB)^{-1}(I-AC)$

$\boxed{y = C - B(C-AB)^{-1}(I-AC)}$

2ΓΒ) $B=\begin{pmatrix} 1 & x & x & x \\ x-1 & 1-x & 0 & 0 \\ x-1 & 0 & 1-x & 0 \\ x-1 & 0 & 0 & 1-x \end{pmatrix}$, $\det B=(x-1)^3\begin{pmatrix} 1 & x & x & x \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}=1\begin{vmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix}$

$1\cdot\begin{vmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}-x\begin{vmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & -1 \end{vmatrix}\Rightarrow -1$ $\quad -x\begin{vmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix}$

Άρα $\det=(x-1)^3$

$(-1-x\cdot y+x\cdot 1-x(-1))=(x-y)^3(-1+x)$

$\Rightarrow \det(B)=(x-y)^3(x-1)=(x-1)^4$

$(x-1)^4\neq 0 \Rightarrow \boxed{x=1}$

Ο πίνακας είναι μιδιάζων για όλες τις τιμές του $x$ εκτός από $\boxed{x=1}$

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Ε.Σ. 4.2) 2γ) $A=\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right)$, $\det A=52$

$A^{(1)}=\begin{pmatrix} 8 & 1 & 2 \\ 1 & -2 & 3 \\ 10 & -7 & 4 \end{pmatrix}\Rightarrow \det A^{(1)}=156$ $\quad x=\frac{|A^{(1)}|}{|A|}=3$

$A^{(2)}=\begin{pmatrix} 1 & 8 & 2 \\ -1 & 1 & 3 \\ 3 & 10 & 4 \end{pmatrix}\Rightarrow \det A^{(2)}=52$ $\quad y=\frac{|A^{(2)}|}{|A|}=1$

$A^{(3)}=\begin{pmatrix} 1 & 1 & 8 \\ -1 & -2 & 1 \\ 3 & -7 & 10 \end{pmatrix}\Rightarrow \det A^{(3)}=104$ $\quad z=\frac{|A^{(3)}|}{|A|}=2$

$\Rightarrow$ μοναδική λύση

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Ε.Σ.) 4.3) 1β) $x_1+2x_2-x_3+2x_4=2$

$2x_1-x_2+2x_3+3x_4=3$

$x_1-2x_3+3x_4=0$

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3δ) $\left.\begin{array}{l} -2y+3z=1 \\ 3x+6y-3z=-2 \\ 6x+6y+3z=5 \end{array}\right\}\Rightarrow\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right]=\left[\begin{array}{ccc|c} 6 & 6 & 3 & 5 \\ 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \end{array}\right]=$

$=\left[\begin{array}{ccc|c} 6 & 6 & 3 & 5 \\ 0 & 6 & -9 & -9 \\ 0 & -2 & 3 & 1 \end{array}\right]=\left[\begin{array}{ccc|c} 6 & 6 & 3 & 5 \\ 0 & 6 & -9 & -9 \\ 0 & 0 & 0 & -6 \end{array}\right]\Rightarrow$ Δ λύσεις

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5) $\left(\begin{array}{ccc|c} 1 & 1 & 3 & a \\ -2 & 5 & 3 & -2a \\ -1 & 8 & 2 & -a \end{array}\right)$ $R_3: 2R_3-R_2$ $\left(\begin{array}{ccc|c} 1 & 1 & 3 & a \\ -2 & 5 & 3 & -2a \\ 0 & 2 & 1 & 0 \end{array}\right)$ $R_2: R_2+2R_1$ $\Rightarrow$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 3 & a \\ 0 & 7 & 9 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right) R_3 : R_3 - \tfrac{1}{7}R_2 \Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 3 & a \\ 0 & 7 & 9 & 0 \\ 0 & 0 & -2/7 & 0 \end{array}\right) \Rightarrow x = a$$
$$\Rightarrow y = 0$$
$$\Rightarrow z = 0$$

6B) $\left.\begin{array}{l} 2x+y+3z=3 \\ x-y-2z=2x \\ x+2y+2z=4x \\ x+y+z=3 \end{array}\right\} \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 1 & -1 & -2 & 2x \\ 1 & 2 & 2 & 4x \\ 1 & 1 & 1 & 3 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -3 & -7 & 4x-3 \\ 1 & 2 & 2 & 4x \\ 1 & 1 & 1 & 3 \end{array}\right] =$

$R_2 : 2R_2 - R_1 \qquad R_3 : 2R_3 - R_1$

$$= \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -3 & -7 & 4x-3 \\ 0 & 3 & 1 & 8x-3 \\ 1 & 1 & 1 & 3 \end{array}\right] \quad R_4 : 2R_4 - R_1 = \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -3 & -7 & 4x-3 \\ 0 & 3 & 1 & 8x-3 \\ 0 & 1 & -1 & 3 \end{array}\right] =$$

$$= \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -3 & -7 & 4x-3 \\ 0 & 0 & -6 & 12x-6 \\ 0 & 1 & -1 & 3 \end{array}\right] = \left[\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 0 & -3 & -7 & 4x-3 \\ 0 & 0 & -6 & 12x-6 \\ 0 & 0 & -10 & 4x+6 \end{array}\right] \quad R_3 = R_3 + R_2$$

$R_4 : 3R_4 + R_2$

$$\begin{array}{l} 2x+y+3z=3 \\ -3y-7z=4x-3 \\ -6z=12x-6 \\ -10z=4x+6 \end{array}$$

$z = -2x+1 \qquad z = -\frac{2}{5}x - \frac{3}{5}$

8) $-2x+1 = -\frac{2x}{5} - \frac{3}{5} \Rightarrow -2x + \frac{2x}{5} = \frac{-3}{5} - 1$

$x = 1$, $y = 2$, $z = -1$

7a) $\left(\begin{array}{ccc|c} 3 & 1 & -1 & 1 \\ 1 & -1 & 2 & 4 \\ 2 & 2 & c & 3 \end{array}\right) R_3 : R_3 - 2R_2 \Rightarrow \left(\begin{array}{ccc|c} 3 & 1 & -1 & 1 \\ 1 & -1 & 2 & 4 \\ 0 & 4 & c-4 & -11 \end{array}\right) R_2 : R_2 - \tfrac{1}{3}R_1 \Rightarrow$

$$\Rightarrow \left(\begin{array}{ccc|c} 3 & 1 & -1 & 1 \\ 0 & -4/3 & 7/3 & 11/3 \\ 0 & 4 & c-4 & -11 \end{array}\right) R_3 : \tfrac{1}{3}R_3 - R_2 \Rightarrow \left(\begin{array}{ccc|c} 3 & 1 & -1 & 1 \\ 0 & -4/3 & 7/3 & 11/3 \\ 0 & 0 & \frac{1}{3}(c-3) & 0 \end{array}\right)$$

$\Rightarrow z = \frac{1}{c-3}$

$\Rightarrow c \neq 3$

E.S. 4.4.) 1B) $\left(\begin{array}{cc} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{array}\right)\left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right) = \left(\begin{array}{cc|cc} \sin\theta & \cos\theta & 1 & 0 \\ 0 & 7/\cos\theta & 1 & \sin\theta/\cos\theta \end{array}\right)$

$R_2 : \frac{\sin\theta}{\cos\theta} R_2 + R_1 \qquad R + S\cos\theta R_2 - \frac{1}{\cos\theta} R_1$

$$\left(\begin{array}{cc|cc} -\sin\theta & 0 & -\frac{\sin\theta}{\cos\theta} & \sin\theta \\ \cos\theta & & & \\ 0 & \frac{7}{\cos\theta} & 1 & \frac{\sin\theta}{\cos\theta} \end{array}\right) \begin{array}{l} R_1 = \frac{-1}{\cos\theta} R_1 + \cos\theta R_2 \\ R_1 = \frac{-1}{\cos\theta} R_1 \\ R_2 = \cos\theta R_2 \end{array} \left(\begin{array}{cc|cc} 1 & 0 & \sin\theta & -\cos\theta\sin\theta \\ 0 & 1 & \cos\theta & \sin\theta \end{array}\right)$$

$= B^{-1}$

2(3ε)) $\left(\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 2 & -5 & -4 & 0 & 1 & 0 \\ 1 & -2 & -3 & 0 & 0 & 1 \end{array}\right) = \left(\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 23 & 10 & 2 & -3 & 0 \\ 0 & -10 & -10 & 0 & 0 & -3 \end{array}\right) R_3 - 3R_1$

$= \left(\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 23 & 10 & 2 & -3 & 0 \\ 0 & 0 & -130 & 20 & -690 & -30 \end{array}\right) = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{147}{3} & 881 & 63 \\ 0 & 1 & 0 & 2123 & -3/23 & -30/299 \\ 0 & 0 & 1 & -2113 & 4/13 & 3/13 \end{array}\right)$

Ex. 5.) 4.5) 2a) $\begin{pmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ 3 & -2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$ $A_{11}=4,\ A_{12}=7,\ A_{13}=13$

$A_{21}=2,\ A_{22}=-7,\ A_{23}=4$

$A_{31}=-7,\ A_{32}=-7,\ A_{33}=7$

$\det A = -21$

$\text{adj}A = \begin{pmatrix} 4 & 27 \\ -7 & -77 \\ -13 & -47 \end{pmatrix} \Rightarrow A^{-1} = \frac{-1}{21}\begin{pmatrix} 4 & 2 & 7 \\ -7 & -7 & 7 \\ -13 & -4 & 7 \end{pmatrix}$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} \Rightarrow \boxed{x = \frac{12}{21}},\ \boxed{y = -1},\ \boxed{z = \frac{-18}{21}}$

Ex. S. 5.2) 5a) $u(7,x,-3)$ / $v(2,5,u) \Rightarrow u \cdot v = 0 \Rightarrow (7,x,-3)(2,5,u) = 0$

$\Rightarrow \boxed{x = 2}$

6γ) $w = (0,1,-2,3)$ / $|w| = \sqrt{0^2+1^2+(-2)^2+3^2} = 4$

7α) $u = (2,3)$ $v = (-2,1)$ / $d(u,v) = [(u_1-v_1)^2+(u_2-v_2)^2]^{1/2} = 2\sqrt{5}$

9α) $u = (0,-6,u)$ $v = (7,3,6)$

$u \cdot v = |u| \cdot |v| \cos\theta \Rightarrow \cos\theta = 0,42 \Rightarrow \boxed{\theta = 83,1}$

10) α) $u(3w+v) = 58$

β) $u(v \cdot w) = 36$

γ) $||u||(2v \cdot w) = 340$

δ) $w \cdot v = (-7,5)(7,7) = -7+35 = 34 + (4,3)$, δεν ορίζεται

ε) $||(u \cdot v)v|| = u \cdot v = 4+21 = 25 = 25(7,7) = (25,175) = \sqrt{25^2+175^2} = 176,78$

στ) δεν ορίζεται

ζ) $(5,10) = \sqrt{5^2+10^2} = 5\sqrt{5}$

η) $||u|| + ||v|| = 5+5\sqrt{2}$ $(||u|| = \sqrt{4^2+3^2} = 5$, $||v|| = \sqrt{7^2+7^2} = 5\sqrt{2})$

18) Θεση $u(x,y)$, $v(2,t)$

$(u.v) = (x,y)(2,t) = xz+yt = 0$ *

$d^2 = (x-z)^2+(y-t^2) \Rightarrow$

$||w|| = \sqrt{x^2+y^2} = 2 = ||v||^2$

$||v|| = \sqrt{z^2+t^2} = 1 = ||v||^2$

$x^2 - 2xz + 2z + y^2 - 2yz + t^2$ / $\|u\|^2 + \|v\|^2 = x^2+y^2+z^2+t^2$

$d^2 = x^2 + 2z + y^2 + t^2$ $\quad x^2+y^2+z^2+t^2 = 5$

$d^2 = 5 \Rightarrow d = \sqrt{5}$ $\quad * \Rightarrow x^2 = -yt$

$2x^2 = -2yt$

Ex. 5) 4) $u = (3,-2,7)$ $\quad v = (7,3,-2)$ $\quad w = (0,0,-5)$

5.3) a) $u \times v = \begin{pmatrix} i & j & k \\ 3 & -2 & 7 \\ 7 & 3 & -2 \end{pmatrix} = (-7,-2,11)$

b) $v \times u = \begin{pmatrix} i & j & k \\ 7 & 3 & -2 \\ 3 & -2 & 7 \end{pmatrix} = -(u \times v) = (7,2,-11)$

g) ~~$u \times (v \times w)$~~ $(u \times w) + (v \times w) = \begin{pmatrix} i & j & k \\ 3 & -2 & 7 \\ 0 & 0 & -5 \end{pmatrix} + \begin{pmatrix} i & j & k \\ 7 & 3 & -2 \\ 0 & 0 & -5 \end{pmatrix} = (25,0,0)$

d) $u \times (v \times w) = \begin{pmatrix} i & j & k \\ 3 & -2 & 7 \\ 15 & 15 & 0 \end{pmatrix} = (-15,-15,15)$

e) $v \times (2v+u) = v \times (5,4,-3) = \begin{pmatrix} i & j & k \\ 7 & 3 & -2 \\ 5 & 4 & -3 \end{pmatrix} = (-7,-7,-11)$

6) $u = (3,-2,7)$ $\quad v = (7,3,-2)$ $\quad w = (0,0,-5)$

a) $u(v \times w) = (3,-2,7)\begin{pmatrix} i & j & k \\ 7 & 3 & -2 \\ 0 & 0 & -5 \end{pmatrix} = -75$

b) $(v \times u)w = -(v \times u)w = 55$ $\quad$ g) $(u \times w)_v = -55$

d) $(u+w)(v \times w) = -75$ $\quad$ e) $(v+w)(v \times w) = 30$

9) a) $P_1(2,7,0)$ $\quad P_2(-1,2,1)$ $\quad P_3(0,-2,0)$

$u = \vec{P_1P_2} = (-3,7,7)$ $\quad v = \vec{P_1P_3} = (-2,3,0)$

$u \times v = \begin{pmatrix} i & j & k \\ -3 & 7 & 7 \\ -2 & -3 & 0 \end{pmatrix} = (3,-2,11)$

$A = \|u \times v\| = \sqrt{194}$

11) a) $(u+w)(u \times w) + u(\vec{u} \times \vec{w}) + w(\vec{u} \times \vec{w}) =$

$= \begin{vmatrix} u_1 & u_2 & u_3 \\ u_1 & u_2 & u_3 \\ w_1 & w_2 & w_3 \end{vmatrix} + \begin{vmatrix} w_1 & w_2 & w_3 \\ w_1 & w_2 & w_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$

11β) $(u+v)\times(u-2v)=(u+v)\times u-(2v)\times(u+v)$

$u+u^{-1}+u\times v-2v\times u-2v\times v$

$u\times v-2v\times u-2v\times v$

$u\times v-2(v\times u)-2(u\times v)=u\times v+2(u\times v)=3(v\times u)$

γ) $(2u-3v)\times(x+2u)=2u\times v-3v\times v+2u\overset{0}{\times}2u-3v\times 2u$

$2u\times v-3v\times 2u$

$2u\times v+6(v\times u)=8(u\times v)$

Εx. Set: 6.7

5α) $\begin{pmatrix}0&0&a\\0&b&0\\c&0&0\end{pmatrix}$ i) ω μη κενό π.χ. $\begin{pmatrix}0&0&v_1\\0&v_2&0\\v_3&0&0\end{pmatrix}$

ii) $\begin{pmatrix}0&0&u_1\\0&u_2&0\\u_3&0&0\end{pmatrix}$, $V=\begin{pmatrix}0&0&v_1\\0&v_2&0\\v_3&0&0\end{pmatrix}$, $U+V=\begin{pmatrix}0&0&u+v_1\\0&u_2+v_2&0\\u_3+v_3&0&0\end{pmatrix}\Rightarrow$ U+V∈ω

iii) $uκ=\begin{pmatrix}0&0&u_1κ\\0&u_2κ&0\\u_3κ&0&0\end{pmatrix}\Rightarrow uκ\in\omega$

i, ii, iii ⇒ ω υπόχωρο $R^{3\times3}$

Εx. 5, 6.2) 4)

α) $\{(2,4,-1),(2,-4,10)\}$ ⇒ linear indipended

β) $\{(4,-3,0),(2,5,-1),(3,1,7)\}$ L, 3 διανυσμ στο $R^3$

γ) $\{(7,-2,0),(5,3,2),(7,6,-1),(-2,7,0)\}$

⇒ 4 διανυσμ. στο $R^3$ ⇒ εξαρτημένο

8) $(2,5,-1)=\int_1(3,1,1)+\int_2(4,-3,0)$

$2=3\int_1+4\int_2$ (3)

$5=\int_1-3\int_2$ (2)

$-1=\int_1$ (7)

$u=(1,7,\int)$, $v=(7,\int,7)$, $\omega=(\int,1,7)$

$u=\kappa_1 v+\kappa_2\omega\Rightarrow(1,7,\int)=\kappa_1(7,\int,7)+\kappa_2(\int,7,7)$

(1) ⇒ $7=\kappa_1 7+\kappa_2\int$

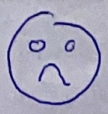
(2) ⇒ $7=\kappa_1\int+\kappa_2$ (1)=(2) ⇒ $\kappa_1+\kappa_2\int=\kappa_1\int+\kappa_2\Rightarrow\kappa_1=\kappa_2$ (4)

(3) ⇒ $\int=\kappa_1+\kappa_2$ (3)=(4) ⇒ $\int=2\kappa$ (5)

(2)=(5) ⇒ $\kappa_3=\frac{-7+\sqrt{5}}{2}$, $\kappa_4=\frac{-1-\sqrt{5}}{2}$

14)

14)...




Ex. Set 6.3) 4) $M_1=\begin{pmatrix}1&0\\0&0\end{pmatrix}$, $M_2=\begin{pmatrix}0&1\\0&0\end{pmatrix}$, $M_3=\begin{pmatrix}0&0\\1&0\end{pmatrix}$, $M_4=\begin{pmatrix}0&0\\0&1\end{pmatrix}$

$\begin{pmatrix}1&0\\0&0\end{pmatrix}=\lambda_1\begin{pmatrix}0&1\\0&0\end{pmatrix}+\lambda_2\begin{pmatrix}0&1\\0&0\end{pmatrix}+\lambda_3\begin{pmatrix}0&0\\0&1\end{pmatrix}$

$1=0\lambda_1+0\lambda_2+0\lambda_3$, $1=0$ (μη δυνατό)

$\Rightarrow$ δεν είναι βάση για $R^{2\times2}$

8β) $-x+y=0$, $-2=x-y$

$\begin{pmatrix}1&-1&1\\-2&2&-1\\-1&1&0\end{pmatrix}\xrightarrow{R_2:R_2+2R_1}\begin{pmatrix}1&-1&1\\0&0&1\\-1&1&0\end{pmatrix}\xrightarrow{R_3\to R_3+R_1}\begin{pmatrix}1&-1&1\\0&0&1\\0&0&1\end{pmatrix}\xrightarrow{R_3\to R_3-R_2}\begin{pmatrix}1&-1&1\\0&0&1\\0&0&0\end{pmatrix}$

$\Rightarrow \begin{matrix}x-y+z=0\\z=0\end{matrix}\Big\}\ x-y=0,\ x=y$ $\quad (x,y,z)=(y,y,0)=y(1,1,0)$

διάσταση $=1$

Ex set 6.5) $A=\begin{pmatrix}1&1&2\\1&0&1\\2&1&3\end{pmatrix}\Rightarrow\begin{pmatrix}1&1&2\\2&1&3\\2&1&3\end{pmatrix}\Rightarrow\begin{pmatrix}1&1&2\\2&1&5\\0&0&0\end{pmatrix}$

$\left\{\begin{pmatrix}1\\2\\0\end{pmatrix},\begin{pmatrix}1\\1\\0\end{pmatrix}\right\}$ $Ax=b=\begin{pmatrix}1\\0\\-2\end{pmatrix}\Big\}$ είναι αφού $b\in R^{3\times1}$, $\text{rank}(A)=3$, άρα είναι συνβιβαστό

12α) $\begin{matrix}2x+3y+z=3\\-x-y=2\\x+2y+z=5\end{matrix}\Big\}\xrightarrow{R_1\leftrightarrow R_2}\left(\begin{array}{ccc|c}2&3&1&3\\-1&-1&0&2\\1&2&1&5\end{array}\right)\xrightarrow{R_1\to-R_1}\left(\begin{array}{ccc|c}1&1&0&-2\\2&3&1&3\\1&2&1&5\end{array}\right)\xrightarrow{\substack{R_2\to R_2-2R_1\\R_3\to R_3-R_1}}\left(\begin{array}{ccc|c}1&1&0&-2\\0&1&1&7\\0&1&1&7\end{array}\right)$

$\xrightarrow{R_3\to R_3-R_2}\left(\begin{array}{ccc|c}1&1&0&-2\\0&1&1&7\\0&0&0&0\end{array}\right)$ $\quad \begin{matrix}x+y=-2\\y+z=7\end{matrix}\Big\}\Rightarrow$ υπάρχουν απείρες λύσεις

Ex Set 8.1) ③α) για το $A$: $p(\lambda)=\det(A-\lambda I_3)=0$

$A-\lambda I_3=\begin{pmatrix}-1&2&2\\2&2&2\\-3&-6&-6\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}-1-\lambda&2&2\\2&2-\lambda&2\\-3&-6&-6-\lambda\end{pmatrix}$

Άρα $p(\lambda)=\det(A-\lambda I_3)=0\Rightarrow\lambda(\lambda^2+5\lambda+6)=0$

$\Rightarrow\lambda_1=0,\ \lambda_2=-2,\ \lambda_3=-3$

για το $B$: $p(\lambda)=\det(B-\lambda I_3)=0$

$B-\lambda I_3=\begin{pmatrix}4&0&-4\\1&2&0\\1&0&0\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}4-\lambda&0&-4\\1&2-\lambda&0\\1&0&-\lambda\end{pmatrix}$

/10

Αρα $p(\lambda) = \det(B - \lambda I_3) = 0$

$\Rightarrow (4-\lambda)(2-\lambda)(-\lambda) - (-4)(2-\lambda)(7) = 0 \Rightarrow \lambda_{1,2,3} = 2$

5) $A = \begin{pmatrix} a_1 & a_2 & a_3 \\ 0 & a_4 & a_5 \\ 0 & 0 & a_6 \end{pmatrix} \Rightarrow \begin{pmatrix} a_1-\lambda & a_2 & a_3 \\ 0 & a_4-\lambda & a_5 \\ 0 & 0 & a_6-\lambda \end{pmatrix}$ $\det(A) = (a_1+\lambda)(a_4-\lambda)(a_6-\lambda)$

όπου $\lambda_1 = a_1, \lambda_2 = a_4, \lambda_3 = a_6$

6) $A = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 4-3 & 6-6 \\ -2+2 & -3+4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

για $A^5 = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} \Rightarrow$ αρα έχει τα ίδια διανύσματα με την (2)

Μηγαδικοί 1.3) 4) α) $z_1$: $Re(z_1) = 3$, $Im(z_1) = 4/-2+I_3$

$z_2$: $R_2(z_2) = -2$, $Im(z_2) = 3$

$z_3$: $R_3(z_3) = 0$, $Im(z_3) = -2$

$z_4$: $Re(z_4) = 4$, $Im(z_4) = 0$

δ) v) $5z_1 - 2z_2 = 19 + 8 \cdot 14$

xi) $z_3 \cdot \bar{z}_3 = (-8_2)(+8_2) - 4$

xiii) $z_2/z_1 = \frac{-2+8_3}{3+8_4} = \frac{-6}{25} + \frac{5+3}{25}$

Εx. Sct) 3α) $z = r(\cos\theta + \sin\theta)$, $r = 2$, $\theta = 45^\circ = \frac{\pi}{4}$ α) $z_1 = 2 < 45^\circ$, $z = x + y$

$\boxed{z_1 = \sqrt{2} + \sqrt{2}}$

$z_1 = 2(\cos 45^\circ + \sin 45^\circ)$, $\cos 45^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$

$z_1 = 2\left|\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right| = \sqrt{2} + \sqrt{2}$

4) $z_1 = 2 - \sqrt{2}$, $z_2 = 2 < \pi/6$, $z_3 = 2e^{-5(\pi/3)}$, $z_4 = -2e^{(\pi/4)}$

$z_2 = 2\cos(\pi/6) + \sqrt{2}\sin(\pi/6) = \sqrt{3} + \sqrt{}$

$z_3 = 2\cos(-\pi/3) + \sqrt{2}\sin(-\pi/3) = -1 - \sqrt{3}$

| α) i) | (ii) | iii) | iv) | v) $Avg(z_1) = 3/5$ |
|---|---|---|---|---|
| $Re(z_1) = 2$ | $Im(z_1) = -2$ | $\bar{z}_1 = 2 + 2\sqrt{2}$ | $\lvert z_1\rvert = v_1 = 8$ | $= \pi/6$ |
| $= \sqrt{3}$ | $= 1$ | $= \sqrt{3} - 5$ | $= 2$ | $= \pi/3$ |
| $= -1$ | $= -\sqrt{3}$ | $= -7 + \sqrt{3}$ | $= 2$ | $= \pi/4$ |
| $= \sqrt{2}$ | $= -\sqrt{2}$ | $= -\sqrt{2} + \sqrt{2}$ | $= -2$ | |

vi) $Avg(\bar{z}_1) = 45$

$= -\pi/6$

$= \pi/3$

$= -\pi/4$

4B) i) $z_1 z_2 = (2+2\sqrt{3}) + j(2-2\sqrt{3})$

ii) $(z_2 z_2) z_4 = (2\sqrt{3}+j^2)(-\sqrt{2}-j\sqrt{2}) = (2\sqrt{2}-2\sqrt{6}) - j(2\sqrt{6}+2\sqrt{2})$

iii) $z_1 z_2 \cdot z_3 = 8 - j8$

iv) $z_1 \cdot \bar{z}_3 = (\sqrt{3}+j)(-1+j\sqrt{3}) = -2\sqrt{3} + j2$

v) $\frac{z_1}{z_3} = \frac{2-j2}{-1-j\sqrt{3}} = \frac{(2\sqrt{3}-2) + j(2\sqrt{3}+2)}{1+\sqrt{3}}$

vi) $\frac{z_1 z_2}{z_3} = \frac{(2\sqrt{3}+j2\sqrt{3}-j2\sqrt{3}+2)\cdot(-1+j\sqrt{3})}{(-1-j\sqrt{3})(-1+j\sqrt{3})}$

7B) $z_2 = \frac{(7-j3)^3}{(-2-j4)^2}$ $\quad 7-j3 = r_1 e^{j\theta_1}$ $\quad r_1 = \sqrt{7^2+(-3)^2} = \sqrt{58}$ $\quad \theta_1 = \sin^{-1}(\frac{-3}{7}) \simeq -23,2^\circ$

$(7-j3) = (\sqrt{58})^3 e^{j3\theta_1} = 58^{3/2} e^{\theta(-1,21)}$

$-2-j4 = r_2 e^{j\theta_2}$

$r_2 = \sqrt{\tan(-2)^2+(4)^2} = \sqrt{4+7} = \sqrt{20}$ $\quad \theta_2 \simeq -116,57^\circ = -2,034$ rad

$\theta_2 = \tan^{-1}(\frac{-4}{-2}) = \tan^{-1}(2) = 63,43^6$ $\quad (-2-j4)^2 (20) e^{i2\theta_2} = 20 e^{j(-4,06)}$

* otro cuadrante

2.1) 1) a) $f(z) = (\bar{z}+2iz)^2$, $f(2-j)$, $f(7+3i)$

$f(z) = [(x-iy)+2i(x+yi)]^2 = u(x,y) + iv(x,y)$

$f(2-j) = [(2+i)+2i(2-3)]^2 = -9 + 40i$

$f(7+3i) = [(7-3i)+2i(7+3i)]^2 = -24 - 10i$

3) a) $\rho < \varphi$ $\quad f(z) = \bar{z} - z_2 = (x-iy) - 2(x+iy) = -x - i3y$

$r = x^2+y^2$, $\varphi = \tan^{-1}(y/x)$ $\quad (x^2+y^2) < \tan^{-1}(y/x)$

4B) $D_2 = (1-i) = \{2i \mid |z-1+i| < 2\}$

$w = (3-4i)z - 2 - 2i$

$w = u+iv = f(z) \Rightarrow f^{-1}(w) = z = x+iy \in D_2(1-i)$

$\text{ap}\alpha \Rightarrow w = (3-4i)z \Rightarrow z_1 - z \Rightarrow z = \frac{w-z+2i}{3-4i} \in D(1-i)$

$\left|\frac{w+z+2i}{3-4i} - 1+i\right| z \Rightarrow |w+3+9i| < 5$, $\text{ap}\alpha$ $w \in D_2(-3-9i)$

2.2) 1) a) $\lim_{z \to -i} [z^2 - 3z(1-2) + z^{-2}] = \lim_{(x,y)\to(0,-1)} [u(x,y)] = \lim_{(x,y)\to(0,-1)} 3x^2 - y^2 - 3x = -1$

$\lim_{(x,y)\to(0,-1)} [v(x,y)] = 3$

2) B) $w = (z^4 - 3z^2 + 7)^4 (4+3iz)^3$, $z_i$

$\lim_{(x,y)\to(0,27)} (z^4 - 3z^2 + 7)^4 (4+3iz)^3 =$

2.4) 1a) $\int_1^2 (2t-3i)^3 dt = \int_1^2 (2t-3i)^2(2t-3i)\,dt =$

$= \left[t^4 - 4it^3 + 9t^2 - 4t^3 i - 18t^2 - \frac{27i}{12}t\right]_1^2 = 12 - i\,83$

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συνέχεια 3.7) 2δ) $C_{22} = (-1)^{2+2} \times \begin{vmatrix} 2 & -7 \\ -7 & 7 \end{vmatrix} = 7 \times 2\cdot 7 - (-2\cdot(-7)) = 7$

$C_{23} = (-1)^5 \times \begin{vmatrix} 2 & 3 \\ -7 & 2 \end{vmatrix} = -7(2\cdot 2 - 3(-7)) = -7(4+3) = -7$

$C_{31} = (-1)^4 \cdot \begin{vmatrix} 3 & -7 \\ 7 & 7 \end{vmatrix} = 7(3\cdot 7 - (-7)7) = 4$

$C_{33} = (-1)^6 \begin{vmatrix} 2 & 3 \\ 0 & 7 \end{vmatrix} = 7(2\cdot 7 - 3\cdot 0) = 2$

$C_{32} = (-1)^5 \begin{vmatrix} 2 & -7 \\ 0 & 7 \end{vmatrix} = -7(2\cdot 7 - 0\cdot(-7)) = -2$

$adj(D) = \begin{bmatrix} -7 & -5 & 4 \\ -7 & 7 & -2 \\ 7 & -7 & 2 \end{bmatrix}$, $D^{-7} = \frac{7}{-6} \times adj(D)$

$D^{-7} = \begin{bmatrix} 7/6 & 5/6 & -2/3 \\ 7/6 & -7/6 & 7/3 \\ -7/6 & 7/6 & -7/3 \end{bmatrix}$

---

συνέχεια 7.4) 7β) $\cdots z_2 = \frac{58^{3/2} e^{i(-7,2\,\pi 47)}}{20 e^{i(-4,0680)}} = \frac{58^{3/2}}{20} \cdot e^{i(2,8)}$

$z_2 = \frac{58^{3/2}}{20} e^{i(2,8)}$

---

2.3) 7a) $\sum_{x=0}^{0} \frac{n(n+7)}{(4+3i)^n} i$ $\quad R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+7}}\right| \Rightarrow a_n = \frac{n(n+7)}{(4+3i)}$

$\left|\frac{a_{n+2}}{a_n}\right| = \frac{(n+7)(n+2)}{(4+3i)} \cdot \frac{(n+3i)^3}{7(n+3)} = \left|\frac{(n+2)}{n} \cdot \frac{1}{4+3i}\right|$

αν $n \to 0$, $\frac{7+2}{n} \to 7$ $\left|\frac{a_{n+7}}{a_n}\right| \to \frac{7}{v+5i} = \frac{7}{5}$

$R = \lim_{n\to+\infty} \left|\frac{a_n}{a_{n+7}}\right| = 5$ $\quad f(z) = \sum_{n=7}^{\infty} \frac{n(n+7)}{(4+3i)^4} \Rightarrow h_2^{-7}$

$= \sum_{n=7}^{\infty} \frac{n^2(n+7)}{(4+3i)} \Rightarrow z^{n-7}$

$f(z) = \sum_{n=0}^{\infty} \frac{(n+7)^2(n+2)}{(4+3i)^{n+2}}$

Z

3.1) 7B) $\int_i (z^2 + 3z)\,dz$, $A=2$, $B=2$

$z(t) = 2(1-t) + (2i)t = 2 + 2t + 2it$, $t \in (0,1)$

$dz = (-2+2i)\,dt$

$$\int (z^2+3z)\,dz = \int_0^1 [z(t)^2 + 3z(t)](-2+2i)\,dt$$

$z(t)^2 = (2-2t+2it)^2 = 4 - 8t + 8it + 4t^2 - 8it^2 + 4i^2t^2$

$= 4 - 8t + 2it + 4t^2 - 8it^2 - 4t^2$

$= 4 - 4t + 8it + 0$

$= 4 - 8t + 8it$

$z_2(t) = 3(2-3t+2it) = 6 - 6t + 6it$

$\Rightarrow (z(t))^2 + 3z(t) = (4 - 8t + 8it) + (6 - 6t + 6it)$

$= 10 - 14t + 14it$

Επομένως:

$$\int_0^1 (10 - 14t + 14it)(-2+2i)\,dt$$

$$= \int_0^1 [-20 + 20i - 14t(-2+2i) + 14it(-2+2i)]\,dt$$

$$= \int_0^1 [-20 + 20i + 28t - 28it + 28t^2 + 28it]\,dt$$

$$= \int_0^1 [-20 + 20i + 20i + 0]\,dt = [-20t + 20it + 14t^2]_0^1$$

$= (-20 + 20i + 14) = -6 + 20i$

Επομένως: $\int_i (z^2 + 3z)\,dz = -6 + 20i$

the end...

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